

# Mutations in *LRRK2* Cause Autosomal-Dominant Parkinsonism with Pleomorphic Pathology

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## Summary

We have previously linked families with autosomal-dominant, late-onset parkinsonism to chromosome 12p11.2-q13.1 (*PARK8*). By high-resolution recombination mapping and candidate gene sequencing in 46

families, we have found six disease-segregating mutations (five missense and one putative splice site mutation) in a gene encoding a large, multifunctional protein, *LRRK2* (leucine-rich repeat kinase 2). It belongs to the ROCO protein family and includes a protein kinase domain of the MAPKKK class and several other major functional domains. Within affected carriers of families A and D, six post mortem diagnoses reveal brainstem dopaminergic degeneration accompanied by strikingly diverse pathologies. These include abnormalities consistent with Lewy body Parkinson's disease, diffuse Lewy body disease, nigral degeneration without distinctive histopathology, and progressive supranuclear palsy-like pathology. Clinical diagnoses of Parkinsonism with dementia or amyotrophy or both, with their associated pathologies, are also noted. Hence, *LRRK2* may be central to the pathogenesis of several major neurodegenerative disorders associated with parkinsonism.

## Introduction

Parkinson's disease (PD) is a common neurodegenerative disorder characterized by a progressive loss of dopaminergic neurons of the substantia nigra, associated with the formation of fibrillar aggregates composed of  $\alpha$ -synuclein and other proteins (Lewy bodies and Lewy neurites). In most cases, PD occurs as a sporadic disease of unknown etiology, but in rare instances, point mutations (Polymeropoulos et al., 1997) or multiplications (Singleton et al., 2003; Chartier-Harlin et al., 2004) of the  $\alpha$ -synuclein gene can cause autosomal-dominant parkinsonism, which resembles sporadic disease in many aspects. Recessive forms of parkinsonism have been recognized that are caused by mutations in the genes for *parkin* (Kitada et al., 1998), *DJ-1* (Bonifati et al., 2003), and *PINK1* (Valente et al., 2004). Additional loci have been mapped on chromosomes 2p (Gasser et al., 1998), 12cen (Funayama et al., 2002), 1q (Hicks et al., 2002), and 2q (Pankratz et al., 2003).

As  $\alpha$ -synuclein aggregation is a pathologic feature common to sporadic and dominantly inherited forms of PD, as well as other neurodegenerative diseases, including dementia with Lewy bodies (DLB) and multiple system atrophy (MSA), those diseases have been collectively called " $\alpha$ -synucleinopathies" (reviewed by Farrer et al., 1999). Other forms of parkinsonism are associated with the accumulation of filaments within neurons and glia, which are composed of the microtubule-associated protein tau. Mutations in this gene (*MAPT*) explain at least a subgroup of families with frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17; Hutton et al., 1998), while sporadic cases with tau pathology most commonly present as progressive supranuclear palsy (PSP) or corticobasal degeneration (CBD) (Dickson, 1999). Based on the putative central role of the tau protein in these diseases, they have been called "tauopathies" (reviewed in Arvanitakis and Wszolek, 2001; Ghetti et al., 2003).

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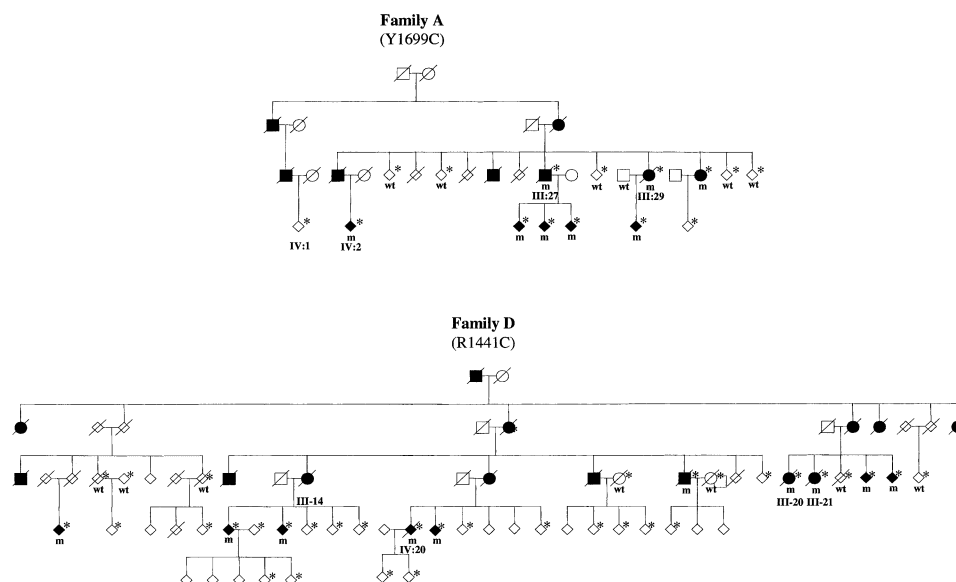


Figure 1. Pedigree Structure of the Two Largest Kindreds with *LRRK2* Mutations

Shown are family A (German-Canadian) and family D (Western Nebraska). Blackened symbols denote affected family members. An asterisk denotes a genotyped individual, with "m" for mutation carriers and "wt" for wild-type *LRRK2*. To maintain confidentiality, genotypes of some unaffected individuals in families A and D are not shown. Additionally, the structure of the pedigrees has been altered, and the genders of individuals in some of the youngest generations have been disguised.

Recently, we have shown that two large families with autosomal-dominant late-onset parkinsonism (families A and D) are linked to the *PARK8* locus on chromosome 12p11.2-q13.1 (OMIM# 607060) (Zimprich et al., 2004), originally mapped in a Japanese family by Funayama et al. (2002). High-resolution recombination mapping, candidate gene sequencing, and the identification of pathogenic mutations within linked families is reported in this article.

## Results

Within our linked families, haplotype analysis of obligate recombinants initially refined a candidate region of 13 Mb, between flanking markers D12S1692 and D12S85. We subsequently sequenced a total of 29 genes in the region, in two patients from each family. Supplemental Table S1 (<http://www.neuron.org/cgi/content/full/44/4/601/DC1/>) lists the 29 genes in order of their genomic position. The primer pairs, sequence variants identified, and sequence chromatograms obtained are available upon request. Missense mutations were identified in a large gene, *LRRK2* (leucine-rich repeat kinase 2), in family A (Y1699C; 5096A>G) and in family D (R1441C; 4321C>T). Both mutations segregated with disease in these two families (Figure 1) and were not found in more than 1000 control individuals or 300 sporadic PD patients.

In family A (Wszolek et al., 1997), 16 individuals were genotyped (8 unaffecteds and 8 affecteds). All affecteds were heterozygous for the mutation, and all unaffecteds aged over 60 were wild-type. Individuals IV:1 and IV:2 (family A) were not included in our initial linkage analysis (Zimprich et al., 2004); both individuals have now been genotyped. Recalculation of the two-point LOD scores

using the mutation as a marker gives maximum LOD scores of 3.78 at  $\theta = 0$  and provides the statistical burden of proof for pathogenicity.

In the second family that was previously linked to the *PARK8* locus, family D (Wszolek et al., 2004), 34 individuals were genotyped (10 affecteds and 24 unaffecteds). All affecteds were heterozygous for the 4321C>T (R1441C) mutation. Of the 24 clinically unaffected individuals who were genotyped, only two were older than 60 years of age and were mutation carriers. These individuals are considered to be at risk and likely presymptomatic, given that the average age of onset in this family is 65 years (Wszolek et al., 2004). To protect the confidentiality of family members, some pedigree symbols and genotypes have been omitted and gender has been disguised in the pedigrees (Figure 1).

To estimate the prevalence of *LRRK2* mutations among PD families, one index patient each from 44 additional families with PD (32 with features consistent with autosomal-dominant PD and 12 affected sib-pairs) were subsequently sequenced. We identified two further missense and one putative splice site mutation, all in families with typical late-onset PD, compatible with a dominant transmission: (I1122V; 3364A>G) in family 21, (I2020T; 6059T>C) in family 32, and (L1114L; 3342A>G) in family 38, which is 6 bp away from the exon/intron border. Affected individuals in an additional family (469) were found to carry the same mutation as family D (R1441C; 4321C>T). These two families are not known to be related, nor do they share haplotypes for the closest flanking microsatellite repeat markers D12S2194, D12S1048, or three newly developed intragenic repeat markers within introns (see Experimental Procedures). This suggests the mutations are either extremely ancient or arose independently. Pedigree structures and clinical

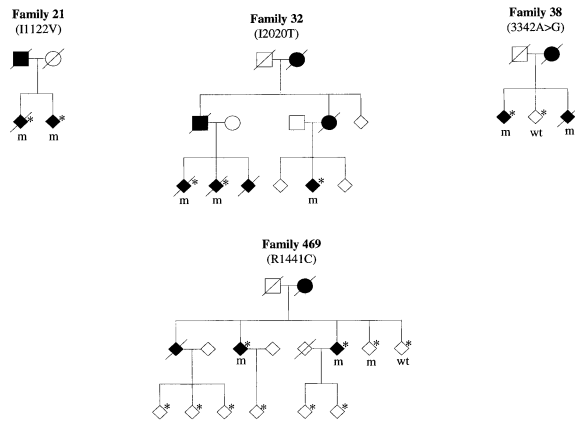


Figure 2. Pedigree Structures of Smaller Families with *LRRK2* Mutations

Blackened symbols denote affected family members. An asterisk denotes a genotyped individual, with "m" for mutation carriers and "wt" for wild-type *LRRK2*. To protect confidentiality, the genotypes of some unaffected individuals of family 469 are not shown. Additionally, the structure of the pedigrees has been altered, and the gender of individuals in the youngest generations has been disguised.

and pathological details of affected patients within these smaller families are illustrated in Figure 2. Clinical and genetic information for all families with *LRRK2* mutations is summarized in Table 1. While smaller families may not demonstrate significant evidence for linkage, the mutations appear to segregate with the disease in all of these families and none were found in controls. Three of the amino acid substitutions (R1441C, Y1699C, and I1122T) are highly conserved across species (Figure 3C).

*LRRK2* was covered by several partial cDNA clones deposited in public data banks. From cross-species sequence alignments, it became clear that the clones were incomplete toward the 5' end. To confirm whether

*LRRK2* is indeed expressed in brain and to establish the complete cDNA sequence, we amplified the whole gene from human brain cDNA (Cat# 7400-1, Clontech, BD Bioscience, San Jose, CA) using overlapping primers predicted by homology searches (see Supplemental Data). The resulting gene spans a genomic region of 144 Kb, with 51 exons encoding 2527 amino acids (Figure 3B). The expression pattern of *LRRK2* was subsequently examined in several tissues. We hybridized human brain and multiple tissue Northern blots with a 1078 bp 3' cDNA probe and found expression in most brain regions, albeit at very low levels. There appeared to be a major transcript of 9 kb and a minor transcript of ~8 kb, which may be explained by the alternative use of polyadenylation [poly(A)] sites. This would be in concordance with two mRNAs, deposited under the accession numbers AL832453 and AL834529, which make use of canonical poly(A) sites (aataaa) at positions 7962 and 8944, respectively. Multiple, very weak bands at lower sizes were found that suggested alternative splicing (data not shown).

Using real-time RT-PCR and RNA isolated from adult and fetal whole brain and from different brain regions, we assessed quantitative gene expression and alternative splicing. We designed primers to generate specific PCR products for exons 1–3, exons 12–13, and exons 32–34, respectively. Within the same tissue or brain region, transcript levels of all three assays showed no significant differences. We found low, but consistent, expression of full-length *LRRK2* in most brain regions, slightly higher in putamen, substantia nigra, and lung (Figure 4). Whether any of these transcripts result in translated, functional protein awaits the generation of specific antibodies. Of note, in adult human brain tissue, exon 6 was constitutively expressed within the full-length mRNA.

*LRRK2* (also referred to as dardarin [see Paisán-Ruiz et al., 2004, published online October 22 and appearing in this issue of *Neuron*]) can be considered a member of the recently defined ROCO protein family (Bosgraaf

Table 1. Clinical and Pathologic Features of Kindreds with *LRRK2* Mutations

| Family                  | A                                                                                                                                                     | D                                                                                                                                                                                                                         | 21         | 32         | 38         | 469        |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|
| Mutation                | Y1699C                                                                                                                                                | R1441C                                                                                                                                                                                                                    | I1122V     | I2020T     | 3342A>G    | R1441C     |
| Mean age at onset       | 53 (35–65)                                                                                                                                            | 65 (48–78)                                                                                                                                                                                                                | 51 (49–53) | 54 (48–59) | 73 (68–77) | 56 (52–60) |
| Mean disease duration   | 13 (5–18)                                                                                                                                             | 13 (4–26)                                                                                                                                                                                                                 | 18 (17,19) | 20 (12–27) | 11 (7,14)  | 23 (21–26) |
| Initial Signs           | RT                                                                                                                                                    | B, RT                                                                                                                                                                                                                     | B          | B          | RT         | RT         |
| Bradykinesia            | +                                                                                                                                                     | +                                                                                                                                                                                                                         | +          | +          | +          | +          |
| Rigidity                | +                                                                                                                                                     | +                                                                                                                                                                                                                         | +          | +          | +          | +          |
| Resting tremor          | +                                                                                                                                                     | +                                                                                                                                                                                                                         | +          | +          | +          | +          |
| Postural instability    | +                                                                                                                                                     | +                                                                                                                                                                                                                         | +          | +          | -          | +          |
| Response to levodopa    | +                                                                                                                                                     | +                                                                                                                                                                                                                         | +          | +          | +          | +          |
| Long-term complications | +                                                                                                                                                     | +                                                                                                                                                                                                                         | -          | -          | -          | -          |
| Other features          | Two cases had amyotrophy and two cases had dementia                                                                                                   | -                                                                                                                                                                                                                         | -          | -          | -          | -          |
| Pathologic features     | Two cases had "nonspecific" SN degeneration with ubiquitin-positive neuronal inclusions. One had concurrent AD and one had mild motor neuron disease. | All cases had SN neuronal loss. One case had "nonspecific" SN degeneration with ubiquitin-positive neuronal inclusions. Two cases had Lewy bodies—one brainstem-type and one diffuse-type. One case had PSP-like changes. | NA         | NA         | NA         | NA         |

Abbreviations: B, bradykinesia; R, rigidity; RT, resting tremor; PI, postural instability; SN, substantia nigra; AD, Alzheimer's disease; PSP, progressive supranuclear palsy.

## *LRRK2* (ROCO gene family)

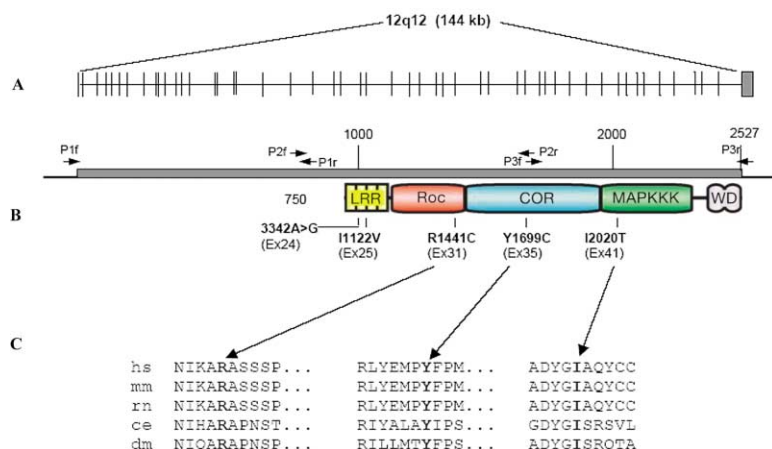


Figure 3. Structure of *LRRK2*

(A) Axon positions.

(B) Schematic drawing of *LRRK2*, with domains, primer positions for cDNA amplification, and positions of mutations.

(C) Protein alignment of three mutations conserved among the following: hs, *Homo sapiens*; mm, *Mus musculus*; rn, *Rattus norvegicus*; ce, *Caenorhabditis elegans*; and dm, *Drosophila melanogaster*.

and Van Haastert, 2003), as predicted *in silico* by the presence of two conserved domains: (1) a Roc (Ras in complex proteins) domain that belongs to the Ras/GTPase superfamily; and (2) a COR domain (C-terminal of Roc) that is also characteristic for this protein family. Ras/GTPase domains are involved in the reorganization of the actin cytoskeleton in response to external stimuli. They also have roles in cell transformation by Ras, as well as in cytokinesis, focal adhesion formation, and the stimulation of stress-activated kinase (Ridley, 2001). Three further conserved domains can be identified: (1) a leucine-rich repeat (LRR), consisting of 12 strands of a 22–28 amino acid motif, present in a tandem array; (2) a tyrosine kinase catalytic domain (TyrKc); and (3) a WD40 domain. Although proteins containing LRRs are associated with diverse functions, such as hormone receptor interactions, enzyme inhibition, cell adhesion, cellular trafficking, splicing, and substrate binding for ubiquitination, a common property involves protein-protein interaction (Kobe and Kajava, 2001). Enzymes with TyrKc domains belong to an extensive family of proteins that share a conserved catalytic core common to both

serine/threonine and tyrosine protein kinases. They exert their function by catalyzing the transfer of the  $\gamma$ -phosphate of ATP to tyrosine residues on protein substrates (Hubbard and Till, 2000). Finally, WD40 domains have been previously implicated in signal transduction, pre-mRNA processing, and cytoskeleton assembly (Smith et al., 1999).

The sequence of this newly identified gene, therefore, suggests multiple functions. As we have found mutations in different functional domains, it is unclear which domain or domains are related to neurodegeneration. It is likely, however, that *LRRK2* may be central to a range of neurodegenerative processes based upon available evidence. First, *LRRK2* mutations appear to be a common cause of autosomal-dominant parkinsonism. In the present study, we identified six independent mutations in 34 families with autosomal-dominant PD. Second, affected individuals with *LRRK2* mutations exhibit strikingly variable pathologic changes (Wszolek et al., 2004), representing aspects of several of the major neurodegenerative diseases (Figure 5).

The common feature of the neuropathology of six af-

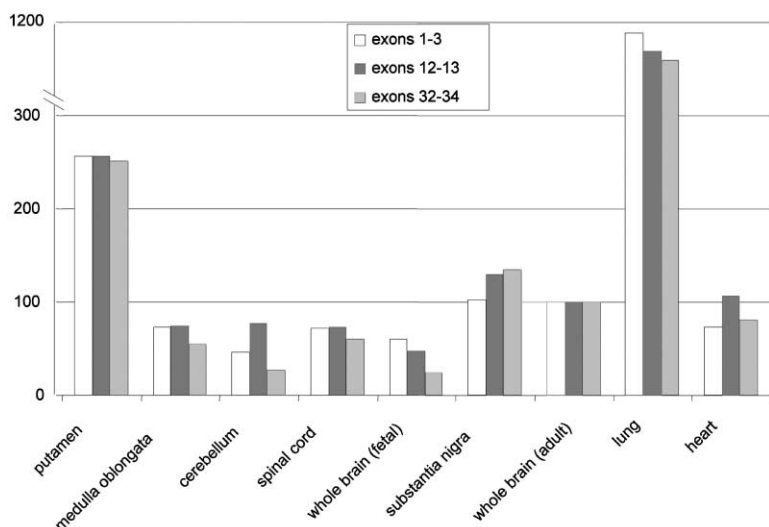
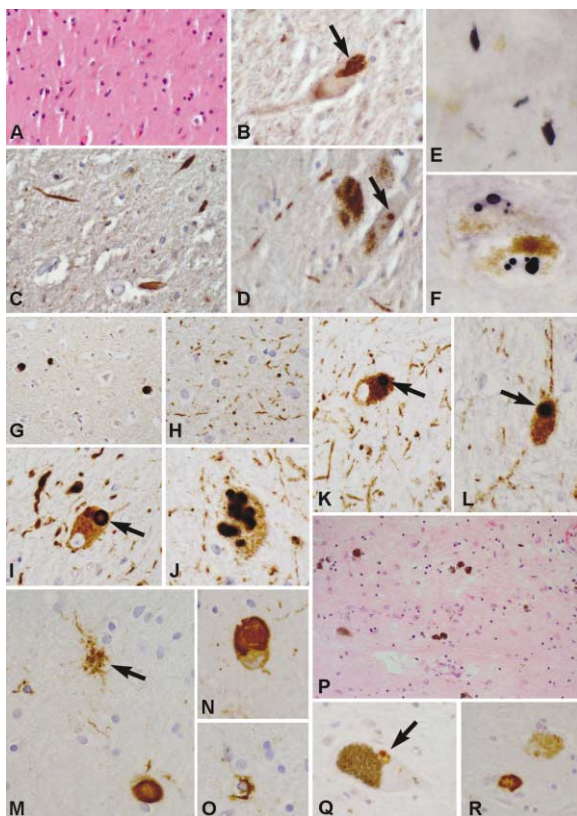


Figure 4. Transcript Levels of *LRRK2* in Different Human Tissues

Quantitative RT-PCR assays show no differential expression of three regions of the *LRRK2* transcripts within the same tissue. Transcript levels are given as ratios of *LRRK2*/h-PBGD, a low-copy housekeeping gene, normalized to adult whole brain expression as 100%. Note the comparably high expression levels of *LRRK2* transcripts in putamen and substantia nigra, brain regions vulnerable to neurodegeneration in PD.



**Figure 5. Substantia Nigra Pathology in Families A and D**  
Summary of substantia nigra pathology in six autopsies from family A (III:29, [A]–[D]; II:27, [E] and [F]) and family D (III:20, [G]–[J]; III:14, [K] and [L]; III:21, [M]–[O]; and IV:20, [P]–[R]). Three cases had nonspecific neuronal loss and gliosis in the substantia nigra (hematoxylin and eosin stain) (A and P) with ubiquitin-immunoreactive neuronal lesions. Ubiquitin immunohistochemistry revealed a range of lesions, including irregular cytoplasmic inclusions (arrow in B) (B and D), curvilinear dystrophic neurites (C and E), granular and foamy axonal spheroids (R), and multiple nuclear inclusions (arrows in D and Q) (D, F, and Q).  $\alpha$ -synuclein immunohistochemistry was positive in two individuals, showing many cortical Lewy bodies (G) and cortical Lewy neurites (H) in one case, as well as typical brainstem-type Lewy bodies (arrow in I, K, and L) or pleomorphic Lewy bodies (J) in the substantia nigra and brainstem nuclei. One case had tau-immunoreactive lesions similar to those seen in PSP, including globe neurofibrillary tangles (M and N), tufted astrocytes (arrow in M), and oligodendroglial coiled bodies (M and O).

affected individuals from families A and D who have come to autopsy is neuronal loss and gliosis in the substantia nigra, which is the likely pathologic substrate of parkinsonism. In three of the individuals (two from family A and one from family D), there was nonspecific loss of dopaminergic neurons, with ubiquitin-immunoreactive cytoplasmic and nuclear inclusions. The latter were similar to Marinesco bodies, while the neuronal cytoplasmic inclusions appear to be novel and await further characterization. Two individuals (both from family D) had Lewy body pathology. In one individual, who died prematurely following a neurosurgical ablation procedure, Lewy bodies were relatively restricted to brainstem nuclei as in typical PD. In the other individual, there was widespread  $\alpha$ -synuclein pathology, including Lewy bodies and Lewy neurites, in a pattern typical of diffuse Lewy body dis-

ease. In the final individual, there were tau-immunoreactive neuronal and glial lesions that morphologically resembled lesions of PSP; however, the density and distribution of the lesions was not diagnostic of PSP. Alzheimer type pathology, which might be expected given the advanced age at autopsy, was detected in several individuals, and one met neuropathologic criteria for Alzheimer's disease. Several individuals in family A had clinical features of motor neuron disease, and there was mild decrease of anterior horn cells, gliosis, and axonal spheroids in these cases (Wszolek et al., 1997).

## Discussion

Although only a small proportion of PD cases are caused by gene mutations with high penetrance, identification of these genes has had a major impact on the understanding of the molecular mechanisms leading to dopaminergic neurodegeneration. Those mechanisms include the aggregation of proteins such as  $\alpha$ -synuclein and tau, deficits in ubiquitin-proteasome protein degradation, or mitochondrial dysfunction and oxidative stress. Until now, it was assumed that these represented separate molecular pathways, all eventually leading to neuronal death, but originating from different initial insults. This view will need to be revised given the present findings.

*LRRK2* appears to be central to the etiology of not just one but several neurodegenerative diseases, including those associated with  $\alpha$ -synuclein and tau pathology and clinical parkinsonism. Previously, it was shown that  $\alpha$ -synuclein and tau pathologies may be closely linked. For example, tau-aggregations have been found in the brains of patients carrying pathogenic A53T mutation in the gene for  $\alpha$ -synuclein (Duda et al., 2002). Tau has also been shown to colocalize with a subset of Lewy bodies in sporadic PD (Ishizawa et al., 2003). Moreover, tau and  $\alpha$ -synuclein have been shown to reciprocally promote fibril formation in vitro (Giasson et al., 2003). The major pathogenic protein aggregating in AD, A $\beta$ , has been shown to promote fibrillization of tau and enhanced formation of neurofibrillary tangles in transgenic animal models (Gotz et al., 2001; Lewis et al., 2001). Interestingly, a genomic region overlapping the PARK8 locus has been identified in a linkage study of familial Alzheimer disease (Scott et al., 2000). Evidence for linkage was derived in a large part from families with at least one member with autopsy-proven diffuse Lewy body disease. Whether this linkage result reflects variants in the *LRRK2* gene remains to be determined.

We hypothesize that the function(s) of *LRRK2* are crucial to the initiation and development of several neurodegenerative disorders. Potentially, *LRRK2* may be responsible for the phosphorylation of both  $\alpha$ -synuclein and tau; its kinase activity could be a key event in the accumulation and aggregation of these unfolded proteins within degenerating neurons.

## Experimental Procedures

### Genetic Analyses

#### DNA Extraction

Genomic DNA from peripheral blood lymphocytes was extracted using standard protocols. Blood was collected after obtaining in-

formed consent from all participating family members. All human studies are approved by institutional review boards of participating institutions.

#### Sequence Analysis

Genomic sequences and annotations were obtained from the National Center for Biotechnology Information (NCBI) (<http://www.ncbi.nlm.nih.gov/>) and University of California, Santa Cruz (UCSC) (<http://genome.ucsc.edu/>). Primers for mutation screening were designed using Primer3 software integrated into script to allow for automated primer design (<http://ihg.gsf.de/ihg/ExonPrimer.html>). We amplified exon sequences and exon-intron boundaries with intronic primers and sequenced them directly by BigDye Terminator Cycle sequencing kit (Applied Biosystems). Primer sequences are provided in the Supplemental Data.

Between markers D12S1692 and D12S85, a total of 29 genes or RNAs were sequenced. These sequences are listed in Supplemental Table S1. In the gene *PPHLN1* (Periplin), we detected an amino acid change (K142E) in one small family of two affecteds. We cannot determine whether this alteration is, in fact, pathogenic, because no further nucleotide change was found, even after extensive mutation analysis in all of the other families investigated.

#### Haplotype Analysis

Haplotypes were constructed by hand using the repeat markers previously used for linkage analysis (Zimprich et al., 2004). Intragenic markers for haplotype analysis for family 469 and family D were established by searching the whole gene for repeat polymorphisms using a "tandem repeat finding program" (<http://c3.biomath.mssm.edu/trf.html>). Polymorphic repeats were found in intron 5 (caa), intron 20 (atct), and intron 29 (ac). Primer sequences for intragenic markers are available from the authors on request.

#### Linkage Analysis

Two point LOD scores were calculated using the MLINK program (V 5.10) in its FASTLINK implementation (V4.1P). Phenocopy rate was set at 0.01 and penetrance for the heterozygous state and homozygous mutation carriers at 0.90. The allele frequency of the disease-causing allele was set at 0.001, as was the frequency of the mutation used as the marker.

#### Screening of Mutations and Polymorphisms

Mutation screening was performed using an ABI 7900 Allelic Detection system. PCR was performed by using Taqman Universal PCR Master Mix and Assays-on-Demand MGB primer and dye-labeled probe sets (Applied Biosystems). For each mutation and polymorphism, at least 1200 control chromosomes from a mixed European descent were screened. In addition, we screened 300 patients with sporadic PD. Genotyping was performed on a MALDI-TOF mass spectrometer (Sequenom MassArray system) using the homogeneous mass-extension (hME) process for producing primer extension products (Tang et al., 1999).

#### Amplification of LRRK2

The complete coding sequence of *LRRK2* was amplified from human brain cDNA by using Marathon-Ready cDNA (BD Biosciences Clontech). Primers were set to amplify three overlapping fragments from exons 1–21 (P1f, P1r), from exons 20–35 (p2f, p2r), and from exons 34–51 (p3f, p3r) (Figure 3). Sequence information was derived from published mRNA of DKFZp434H211. PCR products were run on agarose gel to check length and integrity. All oligonucleotides are available upon request.

#### Northern Blot Analysis

Northern blot analysis was performed according to the manufacturer's protocols (BD Biosciences). For hybridization, we used a *LRRK2* cDNA fragment (bp 6577–7655) corresponding to exon 45–3'UTR.

#### LightCycler Experiments

mRNAs from different human tissues were purchased from BD Biosciences (Clontech BD Sciences, Palo Alto, CA) and were reverse transcribed with the Transcriptor First Strand cDNA Synthesis Kit (Roche Applied Sciences, Mannheim, Germany) according to the manufacturer's protocol. For real-time amplification of *LRRK2*, three specific PCR products spanning exons 1–3, exons 12–13, and exons 32–34 were quantified using the LightCycler Instrument (Roche Applied Sciences). Fluorescence-labeled hybridization probes providing maximum specificity were used for product detection. Calculation of sample concentrations were performed using the fit-point algorithm. The porphobilinogen deaminase (h-PBGD) gene, a low-copy housekeeping gene, was used as an external standard and

quantified using the h-PBGD Housekeeping Gene Set (Roche-Applied Science). Relative transcript levels were calculated as ratios of *LRRK2*/PBGD normalized to adult whole brain adult expression as 100%.

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#### Accession Numbers

The GenBank accession numbers are AY792511 for the human gene (*LRRK2*) and AY792512 for the mouse gene (*Lrrk2*).